Infinite Infimal Convolution Regularization: Concepts, Models and Optimization

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Abstract

Variational methods are a well-established, powerful framework for solving inverse problems. Typically, they combine a discrepancy term that models the forward problem and noise with a regularization term that incorporates prior knowledge. A wide range of regularizers has been proposed in the literature, from classical Tikhonov penalties to modern data-driven and learned functionals. In this context, the infimal convolution of regularization functionals is a well-established tool to balance finitely many such penalties.

In this talk, we introduce infinite infimal convolution as a way to select from a continuum of regularization penalties. This leads to a flexible, data-adaptive model selection mechanism based on an infinite, parameterized family of regularizers. We formulate infinite infimal convolution regularization as a variational problem over Banach-space valued Radon measures and analyze well-posedness and convergence in the setting of linear inverse problems. The framework covers various modeling scenarios, including regularization across smoothness scales and parameterized families of functionals [1]. We further illustrate its use in less direct applications, such as relaxed bilinear inverse problems and the learning of certain classes of regularization functionals. On the optimization side, we show how inverse problems with infinite infimal convolution regularization can be solved using generalized conditional gradient methods [1, 2]. Finally, we explain how the arising subproblems can be treated efficiently via extremal point techniques, yielding practical algorithms with convergence guarantees.

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References

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